Exploratory Data Analysis of Medical Insurance Costs

Dataset: Medical Cost Personal Dataset (Kaggle)

Research Question:

Which factors are associated with medical insurance charges?

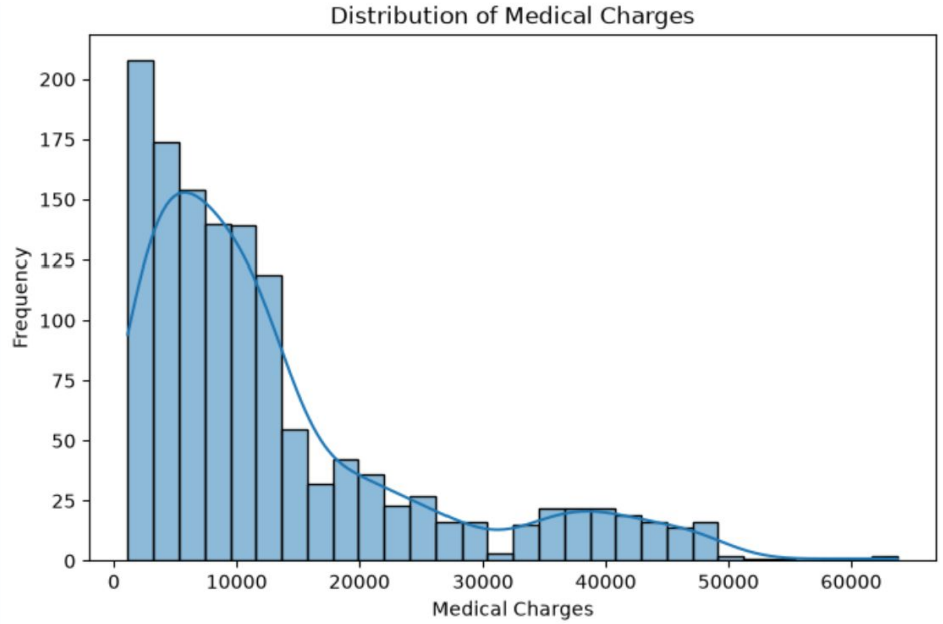
Objective:

To explore the relationships between medical charges and factors such as age, BMI, smoking status, and number of children.

Data Overview & Distribution

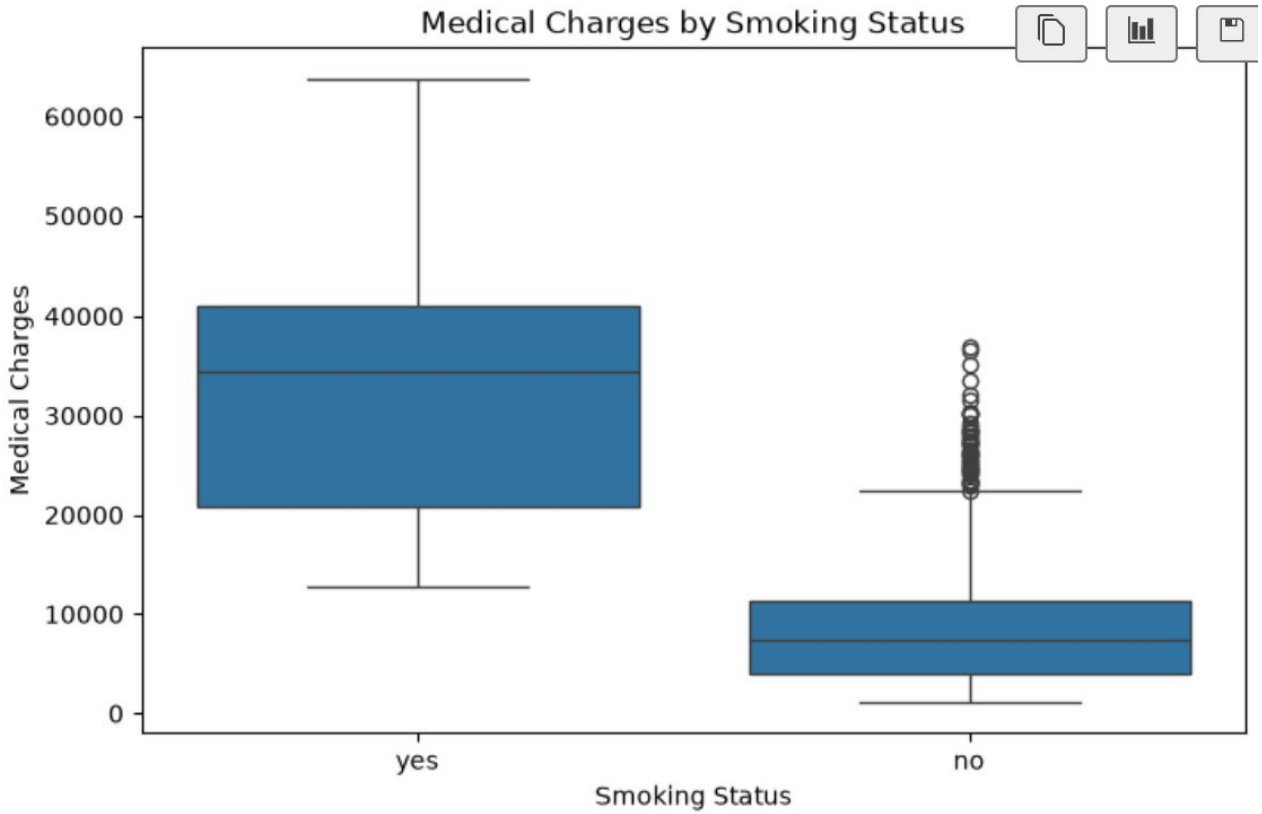
- 1,338 observations and 7 variables
- with no missing values and one duplicated record.
- Medical charges are right-skewed
- The mean charge (13,270), the median (9,382)

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RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   str
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   str
 5   region    1338 non-null   str
 6   charges   1338 non-null   float64
```



Smoking Status & Medical Charges

- mean: smokers 32,050 vs. non-smokers 8,434
- median: smokers 34,456 vs. non-smokers 7,345



BMI & Medical Charges Among Smokers

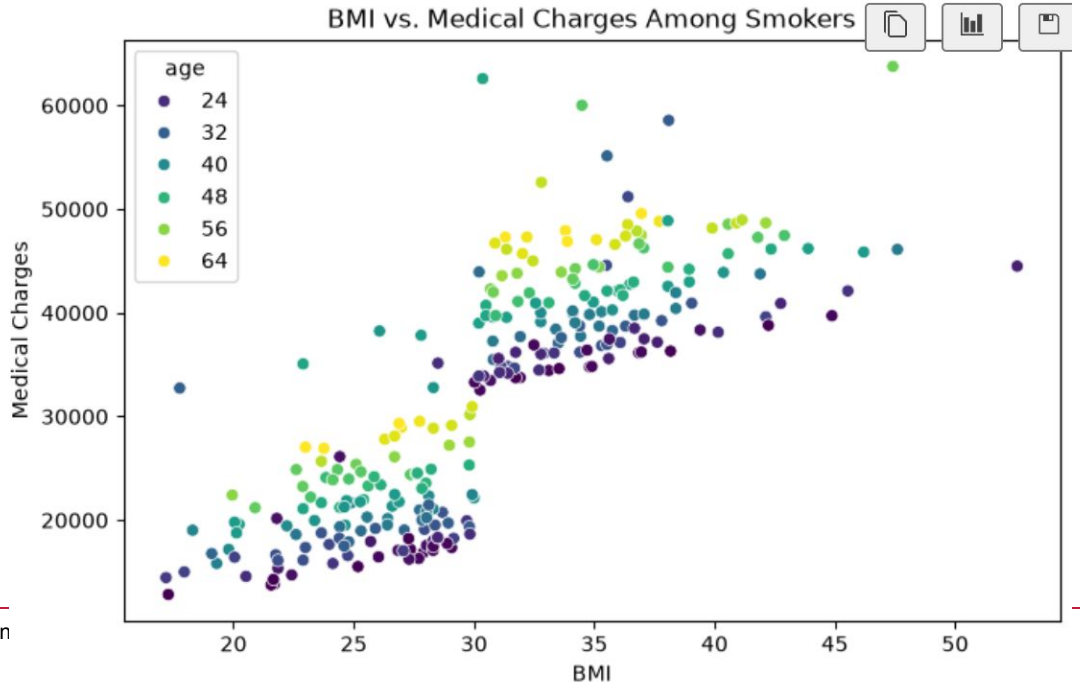
Among smokers,

- average medical charges for those with BMI ≥ 30 (41,558)

  vs. those with BMI < 30 (21,363).

- A noticeable difference around BMI 30.
- Age may also contribute to the charges.

bmi_group	count	mean	median
BMI < 30	129	21363.22	20167.34
BMI >= 30	145	41557.99	40904.20



Correlation Analysis & Conclusion

- the strongest correlation with smoking status.
- weaker positive correlations with age and bmi
- * correlation does not imply causation. * (e.g. covariation, temporal precedence…)

